CSA1717-ARTIFICIAL INTELLIGENCE

PRACTICAL LAB EXPERIMENTS

NIHARIKA PREEJESH (192524506)

BIRD CAN FLY OR NOT

Aim:

To write a Prolog program to determine whether a particular bird can fly.

Objective:

To demonstrate logical inference using facts and rules.

Algorithm:

- **Start.**
- **Define birds.**
- **Define birds that cannot fly.**
- **Define the rule that a bird can fly if it is a bird and is not flightless.**
- **Query the required bird.**
- **Display whether it can fly.**
- **Stop.**

Flowchart:

```
START
  |
  v
Enter Bird
  |
  v
Is it a Bird?
 /       \
No        Yes
 |         |
Cannot    Is it Flightless?
Fly       /       \
       Yes        No
        |          |
     Cannot      Can
      Fly         Fly
        \        /
           STOP
```

Prolog Program:

```
% Bird Flying System

bird(eagle).
bird(parrot).
bird(pigeon).
bird(crow).
bird(ostrich).
bird(penguin).

cannot_fly(ostrich).
cannot_fly(penguin).

can_fly(Bird) :-
   bird(Bird),
   \+ cannot_fly(Bird).
```

Output:

```
78 ?- consult(['c:/Users/Niharika Preejesh/OneDrive/Documents/bird can fly or not.pl']).
true.

79 ?- check_flying(parrot).
parrot can fly.
true
```

Result:

Thus, the Prolog program successfully determines whether a given bird can fly.